Lab: Deep Temporal Models and Hierarchical Inference

> **Learning Goal:** Analyze multi-scale temporal models, trace message passing, and evaluate planning at multiple temporal depths.

Part 1: Constructing a Deep Temporal Model

Exercise: Design a 3-level deep temporal model for driving a car:

Level	Temporal Scale	States Represented	Transition Dynamics
1 (Fast)	~100ms	Steering angle, acceleration, brake pressure	Continuous motor control dynamics
2 (Medium)	~5-30s	Lane position, speed, following distance, maneuver type (cruising, turning, stopping)	Discrete events: lane change, turn, stop
3 (Slow)	~minutes-hours	Route, destination, traffic conditions, trip context	High-level navigation plan

For each level, specify:

1. What observations does it receive (or generate for lower levels)?
2. What predictions does it send to the level below?
3. What prediction errors does it send to the level above?

Now trace what happens during a surprising event: a traffic light turns red unexpectedly.

Write your response here

Part 2: Message Passing Trace

> **Learning Goal:** Trace ascending and descending messages through a concrete scenario.

Exercise: Using the driving model from Part 1, trace the messages during a lane change:

Time t_0: Driver decides to change lanes (Level 3 decision)

1. **Descending L3 → L2**: Prediction: "Begin lane change maneuver"
2. **L2 update**: L2 updates its belief: maneuver = lane_change. This generates...
3. **Descending L2 → L1**: Prediction: "Steering angle should shift by X degrees over Y seconds"
4. **L1 execution**: Motor system generates proprioceptive predictions fulfilled by steering reflexes

Time t_1: A car appears in the blind spot (unexpected!) → Level 1 sensory prediction error

1. **Ascending L1 → L2**: PE: "Something is in the target lane — large visual prediction error"
2. **L2 update**: L2 revises maneuver → abort lane change
3. **Ascending L2 → L3**: PE: "Lane change failed, alternative route needed"
4. **L3 update**: Route plan may need revision

Write a 200-word analysis: How does precision weighting determine whether the ascending PE is strong enough to override the descending prediction?

Write your response here

Part 3: Planning at Multiple Scales

> **Learning Goal:** Analyze how expected free energy operates at different temporal scales.

Exercise: For a student planning their day, compute expected free energy at three levels:

Level 3 (Day plan): Policies: {go to class, skip class, study at home}

- G_3(go to class) = pragmatic value (learns material) + epistemic value (resolves exam uncertainty)
- G_3(skip class) = pragmatic value (rest) + epistemic cost (misses material)

Level 2 (Hour plan, given "go to class"): Policies: {sit front, sit back, participate actively}

- G_2(participate) = pragmatic (better grade) + epistemic (gets feedback on understanding)
- G_2(sit back) = lower engagement cost but lower information gain

Level 1 (Moment plan, given "participate actively"): Policies: {raise hand, type notes, listen}

- G_1(raise hand) = epistemic value (resolve specific confusion)

How do higher-level policies constrain lower-level EFE calculations?

Write your response here

Part 4: Imagination and Counterfactuals

> **Learning Goal:** Analyze how deep temporal models enable mental simulation.

Exercise: Design a counterfactual reasoning scenario:

Scenario: You are deciding between two job offers — Company A (safe, stable, known) and Company B (exciting, risky, unknown).

1. **Generative model forward pass for Company A**: Run the model forward 5 years. What does Level 3 predict? Level 2 (yearly events)? Level 1 (daily experience)?
2. **Forward pass for Company B**: Same exercise. Note the higher uncertainty (wider distributions) at each level.
3. **EFE comparison**: Company A has lower pragmatic uncertainty (known reward) but lower epistemic value (nothing to learn). Company B has higher epistemic value but higher pragmatic uncertainty.
4. **How does the model resolve this?** What role does risk tolerance (precision on C vector) play?

Write your response here

Part 5: Reflection

In 300 words, reflect: Deep temporal models suggest that consciousness may be related to the depth of the temporal hierarchy — deeper models enable richer subjective experience. Do you find this plausible? What aspects of consciousness does it explain? What does it miss?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Model construction	Multi-level temporal architecture
2	Message tracing	Ascending/descending information flow
3	EFE computation	Multi-scale planning
4	Counterfactual reasoning	Mental simulation through forward model
5	Philosophical reflection	Temporal depth and consciousness